

Development of an In-Ear Wearable Device for Continuous Monitoring of Elderly Patients with Dementia: A Preliminary Approach



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Introduction

Globally, more than 55 million people live with dementia, the caregiver's role becomes increasingly complex. Wearable technology can offer valuable support in monitoring this health condition World Health Organization (2025b).

The system project overview, fig.1, is as part of a wider system under development, with two types of device being created: an in-ear device (HowMI) Costa et al. (2023), and a wrist device, each designed to monitor specific physiological parameters and behaviours parameters.

This research continues the development of a wearable device HowMI. Thus, the focus of this work is the modularization of an ear-worn device, HowMI, which can be used both on the ear and on the wrist.

This version of the device, HowMiv2, is intended to continuously monitor physiological signals in elderly people with dementia, including heart rate, blood oxygen saturation, and body temperature. The device communicates via Bluetooth Low Energy with a mobile application that allows its operation and control.

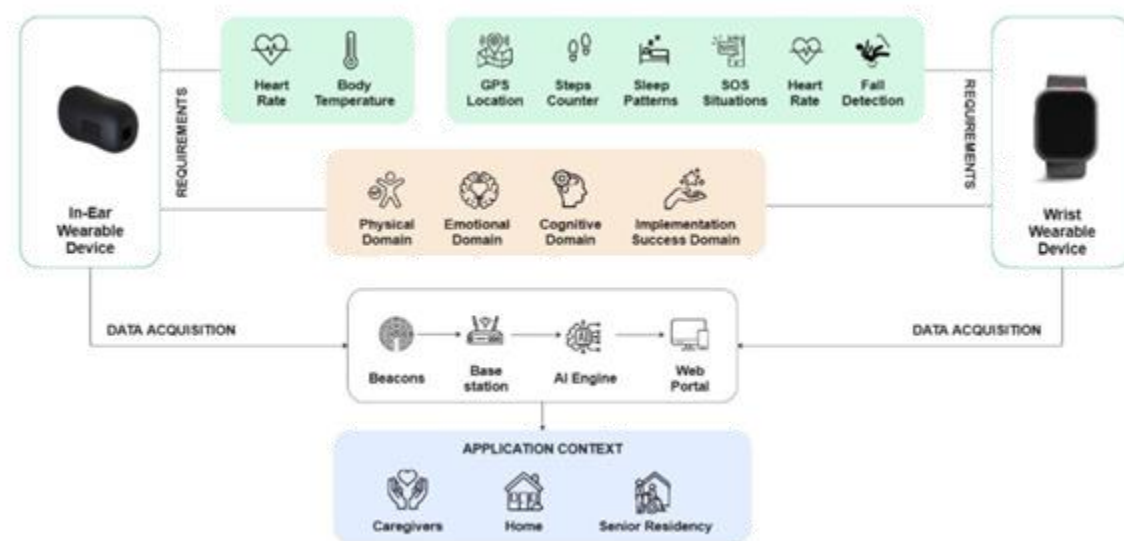


Fig. 1 -Platform for routine measurement and monitoring overview, adapted from (Arantes et al., 2025).

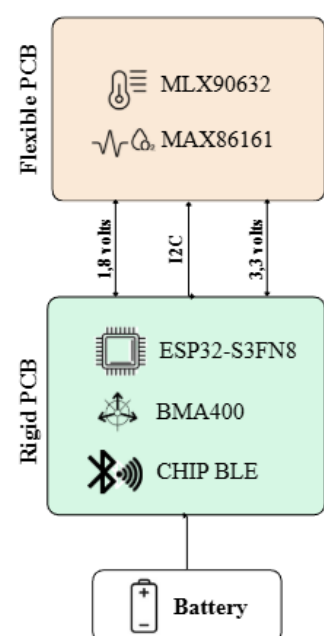


Fig. 2 –Distribution of components in the device hardware.

Materials And Methods

A flex-rigid PCB hardware solution was developed, fig.3, taking into account the anatomical constraints of the ear. The rigid PCB contributes to the mechanical rigidity and structural consolidation of the device, while the flexible PCB adapts to the natural curvature of the ear. Control and processing components are placed on the rigid PCB, whereas the sensing components are integrated into the flexible PCB, fig.2.

At the firmware level, the system is based on a real-time operating system (FreeRTOS). The sensors operate using firmware provided by the manufacturers, in accordance with their respective datasheets.

For device communication and control, a mobile application was developed, enabling switching between two monitoring modes: continuous monitoring and manual monitoring of physiological signals. The application allows the definition of time intervals for sensor measurements, visualisation of historical data with date and time stamps, and the configuration of safe value ranges for alert notifications

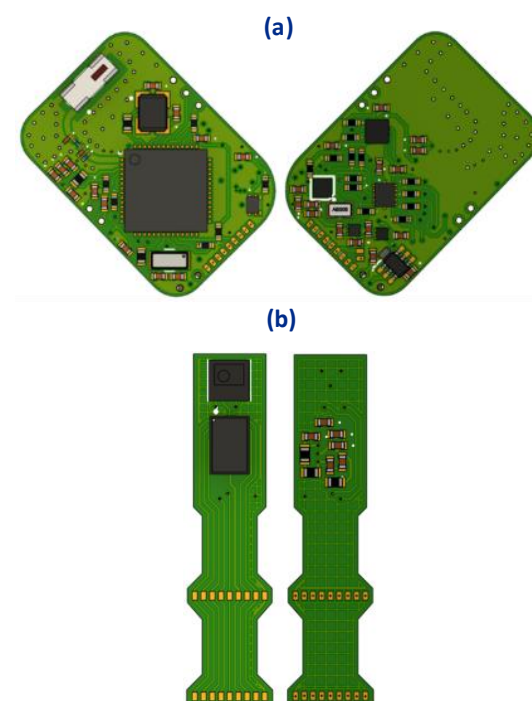


Fig. 3 – Device Hardware (a)-Rigid PCB, (b)-Flexible PCB.

Conclusions And Future Work

Despite these contributions, the current version, fig.4 of the device presents limitations, as it remains at a prototype stage. The energy consumption has not yet been characterised, and a comprehensive validation of the accuracy of the sensor measurements has also not yet been performed.

Future work will focus on several aspects:

- Energy optimisation at both firmware and hardware levels
- Incorporation of artificial intelligence (AI) to improve physiological signal acquisition
- Evaluation of this prototype to assess the feasibility of integrating additional biometric metrics

User testing and clinical trials will be conducted to evaluate the accuracy of the measurements, as well as the comfort, usability, and acceptance of the device among older adults.

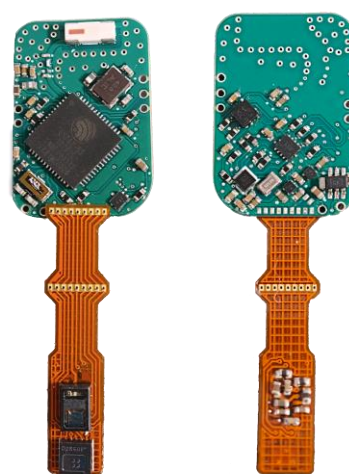


Fig. 4 – HowMiv2 hardware solution.

References

- World Health Organization. (2025b, October 1). Ageing and health.
- Arantes, M., Rocha, I., Carvalho, M., Freitas, A., Soares, J., Pinto, M., Matos, D., Morais, P., & Carvalho, V. (2025). Inclusive Monitoring in Dementia: Progress Towards a Wrist Wearable Based Solution. EAI Endorsed Transactions on Digital Transformation of Industrial Processes, 1(3). <https://doi.org/10.4108/dtip.9843>
- Costa, C., Faria, J. M., Guimarães, D., Matos, D., Moreira, A. H. J., Morais, P., Vilaça, J. L., & Carvalho, V. (2023). A Wearable Monitoring Device for COVID-19 Biometric Symptoms Detection. IRBM, 44(6), 100810. <https://doi.org/10.1016/j.irbm.2023.100810>

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